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SUPERSEDING
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Fed. Spec. DD-G-451a
December 9, 1949

FEDERAL SPECIFICATION

GLASS, PLATE, SHEET, FIGURED (FLOAT, FLAT, FOR GLAZING, CORRUGATED, MIRRORS AND OTHER USES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cut sizes of flat and corrugated glass for glazing and other uses.

1.2 Classification. Glass shall be of the following types, classes, styles, kinds, forms,and qualities, as specified (see 6.2):

Type I - Plate or float, flat.

Class 1 - Transparent.

Quality q1 - Silvering.
q2 - Mirror glazing.
q3 - Glazing.

Class 2 - Heat absorbing and light reducing quality q3.

Style A - Higher light transmission.

Style B - Lower light transmission

Class 3 - Light reducing, tinted, quality q3.

Type II - Sheet, flat.

Class 1 - Transparent.

Quality q4 - AA.
q5 - A.
q6 - B.
q7 - A silvering.
q8 - B silvering.
q9 - Greenhouse.

Class 2 - Heat absorbing, quality q6.

Style A - Higher light transmission.

Style B - Lower light transmission.

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Class 3 - Light reducing, tinted, quality q6

Type III - Rolled, flat and corrugated.

Class 1 - Translucent.

Class 2 - Heat absorbing.

Style A - Higher light transmission.

Style B - Lower light transmission.

Class 3 - Light reducing, tinted.

Kind (classes 1, 2, and 3):

Kind A - flat.

Forms 1 - Wired, polished both sides.
2 - Wired and figured.
3 - Figured.
Quality q10 - Decorative.
q11 - Glazing.
Finish f1 - Figured one side.
f2 - Figured both sides.
4 - Rough surface finishes (rolled rough plate).
Finish f3 - Roudh one side, polished other.
f4 - Rough one side, smooth other.
f5 - Rough both sides.

Mesh (forms 1 and 2):

m1 - Diamond.
m2 - Hexagonal.
m3 - Square.
m4 - Rectangular.
m5 - Parallel strand.
m6 - Special.

Pattern (forms 2 and 3):

p1 - linear, uniform.
p2 - linear, random.
p3 - Geometric, deep.
p4 - Geometric, shallow.
p5 - Random - deep.
p6 - Random - shallow.
p7 - Fluted, large.
p8 - Fluted, small.
p9 - Shallow, uniform, quality q11.
p10 - Spaced ribs, quality q11.
p11 - Special.

Kind B - Corrugated.

Forms 5 - Plain.
6 - Wired, hexagonal mesh.
7 - Special.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

QQ-S-781 - Steel, Strapping, Flat.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment(Civilian Agencies).

(Activities outside the Federal Goverbment may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles,and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., **1616 P St., N. W., Washington, D. C. 20036.)**

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. Glass shall be a fused inorganic substance madefrom a mixture of natural silica sand and two or more alkaline salts such as soda, lime, or potash.

3.2 Types I, II, and III.

3.2.1 Edges. Edges shall be cut, ground, swiped, seamed, round and polished, flat polished, or beveled and mitred, as specified (see 6.2).

3.2.2 Marking. Each light of glass, except when recut, shall be labeled with the manufacturer's name or brand, the quality of the glass, and the nominal thickness designation, unless otherwise specified (see 6.2).

3.3 Type I, plate or float.

3.3.1 Dimensional tolerances,classes 1, 2,and 3. Tolerances for length, width and thickness shall be in accordance with table I (see 6.2).

TABLE I. Dimensional tolerances for plate or float glass, rectangular shapes

Thickness		Tolerance				Qualities
		Thickness plus or minus		Length and width, cut sizes plus or minus		
Inches	Mm.	Inch	Mm.	Inch	Mm.	
1/8	3.2	1/32	0.8	1/16	1.6	q1, q2, q3
3/16	4.8	1/32	.8	1/16	1.6	q1, q2, q3
13/64	5.2	1/32	.8	1/16	1.6	q1, q2, q3
7/32	5.6	1/32	.8	1/16	1.6	q1, q2, q3
1/4	6.4	1/32	0.8	1/16	1.6	q1, q2, q3
5/16	7.9	1/32	.8	5/64	2.0	q3
3/8	9.5	1/32	.8	3/32	2.4	q3
1/2	13	1/32	.8	1/8	3.2	q3
5/8	16	1/16	1.6	5/32	4.0	q3
3/4	19	1/16	1.6	3/16	4.8	q3
7/8	22	1/8	3.2	7/32	5.6	q3
1	25	1/8	3.2	1/4	6.4	q3
1-1/4	32	1/8	3.2	5/16	7.9	q3

3.3.2 Defects in plate and float glass. Defects shall be not greater than those listed in tables II and III for the applicable quality.

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TABLE II. Plate or float glass, type I, class 1, qualities q1 and q2, allowable defects for thicknesses 1/4 inch (6.4mm) or less (i)

Defects	q1 Silvering quality up to 25 square feet (2.3m²)		q2 Mirror glazing quality up to 25 square feet (2.3m²)		q2 Mirror glazing quality over 25 square feet (2.3m²)	
	Central[d]	Outer[d]	Central[d]	Outer[d]	Central[d]	Outer[d]
Gaseous inclusions	1/64 inch[a,g] (0.4mm.)	1/32 inch[a,g] (0.8mm.)	1/32 inch[a,g] (0.8mm.)	3/64 inch[a,g] (1.2mm.)	3/64inch[a,g] 1.2mm.)	1/64 inch[a,g] (1.6mm.)
Open gaseous inclusions	None	1/64 inch[b,g]	None	1/64 inch[a,g] (0.4mm.)	1/64 inch (0.4mm.)	1/32 inch[a,g] (0.8mm
Stones and knots	None	None	None	1/64 inch[b] (0.4 mm.)	1/64 inch[b] (0.4mm.)	1/32 inch[a] (0.8mm.)
Ream, strings and distortion	30°c		30°c		30°c	
Scratches and rubs	Faint[e] <6 inches (15.2cm.)	Light[e] <6 inches (15.2cm.) Faint[e] >6 inches (15.2cm.)	Faint[e] <6 inches (15.2cm.)	Light[e] <6 inches (15.2cm.) Faint[e] >6 inches (15.2cm.)	Light[e] <6 inches (15.2cm.)	Medium[e] <6 inches (15.2cm.) Light[e] >6 inches (15.2cm.)
Crush	None	None	None	None	Light[e] <1/64 inch[b] (0.4 mm.)	Medium[c] <1/32 inch[a] (0.8mm.)
Digs	None	None	None	None	1/64 inch[b] (0.4mm.)	1/32 inch[b] (0.8mm.)
Process surface defects	Faint[h]	Light[h]	Faint[h]	Light[h]	Light[h]	Medium[h]

Notes: (see table III.)

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TABLE III. Plate or float glass, type I, classes 1, 2 and 3, quality q3 allowable defects for thicknesses 1/4 inch (6.4mm.) or less (i)

Defects	q3 Glazing quality					
	Up to 25 square feet ($2.3m^2$)		25 to 75 square feet (2.3 to $7.0m^2$)		Over 72 square feet ($7.0m^2$)	
	Central[d]	Outer[d]	Central[d]	Outer[d]	Central[d]	Outer[d]
Gaseous inclusions	1/16 inch[a,g] (1.6mm.)	3/32 inch[a,g] (2.4mm.)	1/8 inch[a,g] (3.2mm.)	3/16 inch[a,g] (4.8mm.)	1/4 inch[a,g] (6.4mm.)	1/4 inch[a,g] (6.4mm.)
Open gaseous inclusions	3/64 inch[a,g] (1.2mm.)	1/16 inch[a,g] (1.6mm.)	3/64 inch[a,g] (1.2mm.)	1/16 inch[a,g] (1.6mm.)	1/4 inch[a,g] (6.4mm.)	1/4 inch[a,g] (6.4mm.)
Stones and knots	1/64 inch[a] (0.4 mm.)	1/32 inch[a] (0.8mm.)	1/16 inch[a] (1.6mm.)	1/16 inch[a] (1.6mm.)	1/8 inch[a] (3.2mm.)	1/8 inch[a] (3.2mm.)
Ream and strings and distortion	45°[c]		90°[c]		90°[c]	
Scratches and rubs	Medium[e]	Medium[e]	Medium[e]	Heavy[e]	Heavy[e]	Heavy[e]
Crush	Medium[e] <1/16 inch[b] (1/6mm.)	Medium[e] <3/32 inch[b] (2.4mm.)	Medium[e] <1/8 inch[b] (3.2mm.)	Heavy[e] <3/16 inch[a] (4.8mm.)	Heavy[e] <1/4 inch[b] (6.4mm.)	Heavy[e] <1/4 inch[a] (6.4mm.)
Digs	1/16 inch[b] (1.6mm.)	3/32 inch[b] (2.4mm.)	1/8 inch[a] (3.2mm.)	3/16 inch[a] (4.8mm.)	1/4 inch[a] (6.4mm.)	1/4 inch[a] (6.4mm.)
Process surface defects	Medium[h]	Medium[h]	Medium[h]	Heavy[h]	Heavy[h]	Heavy[h]

Notes:

(a) Separated by at least 12 inches (30.5cm.).
(b) Separated by at least 24 inches (61.0cm.).
(c) Vision interference angle (see 4.3.2).
(d) Central and outer areas. The central area is considered to form an oval or circle centered on the light whose axis or diameters do not exceed 80 percent of the overall dimension. The remaining area is considered the outer area.
(e) Intensity - scratches, rubs, and crush. When looking through glass and perpendicular to it using daylight without direct sunlight or with background light suitable for observing each type of defect, the defects shall not be detectable at distances greater than the following, except for heavy intensity (see 4.3.3):

Intensity	Distance
Faint	8 inches (20.3cm.)
Light	3 feet (91.4cm.)
Medium	11 feet (3.35m.)
Heavy	>11 feet (3.35m.)

(f) Gaseous inclusions may be round or elongated. For elongated inclusions maximum size specified above shall be determined by adding the length and width of the inclusion and dividing by two $(l+w)/2$.
(g) For defects of small size or of less intensity, the minimum separation shall be proportionately less. The larger of two defects shall govern the separation. Defects not specifically mentioned shall be compared to the defect they most closely resemble.

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(h) Intensity - process surface defects. When viewed in normal reflected light, the defects are classified as follows: Faint - visible only to the trained eye. Light - just noticeable. Medium - visible as a slight grayish haze. Heavy - readily visible as a cloudy surface.

(i) Glazing quality glass greater than 1/4 inch (6.4mm.) in thickness may contain proportionally more and larger defects.

3.4 Type II, sheet, flat.

3.4.1 Dimensional tolerances, classes 1, 2 and 3. Tolerances for length, width, and thickness shall be in accordance with table IV (see 6.2).

TABLE IV. Dimensional tolerances for sheet glass, rectangular shapes

Designation and thickness		Tolerance, thickness minimum		Tolerance, thickness maximum		Length and width cut sizes plus or minus		Qualities
		Inch	Mm.	Inch	Mm.	Inch	Mm.	
Micro		0.030	0.8	0.042	1.1	1/32	0.8	q4, q5, q6
Lantern slide		.043	1.1	.053	1.3	1/32	.8	q4, q5, q6
Photo		.054	1.4	.069	1.8	1/32	.8	q4, q5, q6
Picture		.070	1.8	.080	2.0	1/32	.8	q4, q5, q6
Single strength		.085	2.2	.101	2.6	1/32	.8	q4, q5, q6, q7, q8
Double strength		.115	2.9	.134	3.4	1/32	.8	q4, q5, q6, q7, q8,q9
3/16 inch	(4.8mm.)	.182	4.6	.205	5.2	1/16	1.6	q4, q5, q6, q7, q8
7/32	(5.6)	.206	5.2	.230	5.8	1/16	1.6	q4, q5, q6, q7, q8
1/4	(6.4)	.236	6.0	.264	6.7	1/16	1.6	q4, q5, q6, q7, q8
11/32	(8.7)	.320	8.1	.355	9.0	1/16	1.6	q6
3/8	(9.5)	.357	9.1	.393	10.0	3/32	2.4	q6
7/16	(11.1)	.396	10.1	.434	11.0	3/32	2.4	q6
1/2	(12.7)	.478	12.1	.522	13.3	1/8	3.2	q6

3.4.2 Defects in sheet glass.

3.4.2.1 Qualities q4, q5, q6, q7, and q8. Defects shall be not greater than those listed in tables V and VI for the applicable quality.

TABLE V. Sheet glass, qualities q4 and q5, allowable defects

Defects	q4 AA quality		q5 A quality	
	Central[d]	Outer[d]	Central[d]	Outer[d]
Gaseous inclusions	1/16 inch[a] (1.6mm.)	1/8 inch[a] (3.2mm.)	1/4 inch[a] (6.4mm.)	1/2 inch[a] (12.7mm.)
Opaque gaseous inclusions	None	None	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)
Knots, dirt, and stones	None	1/32 inch[b] (0.8mm.)	1/32 inch[b] (0.8mm.)	1/16 inch[b] (1.6mm.)
Scratches and rubs	Faint[e]	Faint[e]	Light[e]	Light[e]
Crush	Faint[b,e] <1/16 inch (1.6mm.)	Faint[b,e] <1/8 inch (3.2mm.)	Light[b,e] <1/4 inch (6.4mm.)	Light[b,e] <1/2 inch (12.7mm.)
Digs	1/16 inch[b] (1.6mm.)	1/8 inch[b] (3.2mm.)	1/4 inch[b] (6.35mm.)	1/2 inch[b] (12.7mm.)
Wave, ream, strings, and lines	45°[c]		45°[c]	

Notes: (See table VI.)

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TABLE VI. Sheet glass, qualities q6, q7, and q8, allowable defects

Defects	q6 B quality		q7 A silvering quality		q8 B Silvering quality	
	Central[d]	Outer[d]	Central[d]	Outer[d]	Central[d]	Outer[d]
Gaseous inclusions	1/2 inch[a] (12.7mm.)	3/4 inch[a] (19.0mm.)	1/4 inch[b] (6.4mm.)	1/2 inch[b] (12.7mm.)	1/2 inch[b] 12.7mm.)	3/4 inch[b] (19.0mm.)
Opaque gaseous inclusions	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)	1/4 inch[b] (6.4mm.)
Knots, dirt, and stones	1/16 inch[b] (1.6mm.)	1/8 inch[b] (3.2mm.)	1/32 inch[b] (0.8mm.)	1/32 inch[b] (0.8mm.)	1/32 inch[b] (0.8mm.)	1/32 inch[b] (0.8mm.)
Scratches, and rubs	Medium[e]	Medium[e]	Faint[e]	Faint[e]	Faint[e]	Faint[e]
Crush	Medium[b,e] 1/2 inch (12.7mm.)	Medium[b,e] 3/4 inch (19.0mm.)	Faint[b,e] 1/4 inch (6.4mm.)	Faint[b,e] 1/2 inch (12.7mm.)	Faint[b,e] 1/2 inch (12.7mm.)	Faint[b,e] 3/4 inch (19.0mm.)
Digs	1/2 inch[b] (12.7mm.)	3/4 inch[b] (19.0mm.)	1/4 inch[b] (6.4mm.)	1/2 inch[b] (12.7mm.)	1/2 inch[b] (12.7mm.)	3/4 inch[b] (19.0mm.)
Wave, ream, strings, and lines	90°[c]		45°[c]		90°[c]	

Notes: (a), (b), (c), (e), (g) - See corresponding notes in table III.

(d) The central area is considered to form an oval or circle centered on the light whose axes or diameters do not exceed 50 percent of the over-all dimension. The remaining area is considered the outer area.

3.4.2.2 Quality q9, green house. Glass may contain defects of any size or intensity but shall contain no stones which may cause spontaneous breakage.

3.5 Type III, rolled, flat and corrugated.

3.5.1 Kind A, classes 1, 2, and 3, flat.

3.5.1.1 Forms 1 and 2 wired.

3.5.1.1.1 Thickness and dimensional tolerances. The allowable tolerances for rectangular shapes shall be plus or minus 1/16 inch (1.6mm.) for each 1/8 inch (3.2mm.) of nominal thickness. Thickness, and tolerance on thickness shall be in accordance with table VII (see 6.2).

TABLE VII. Thickness and tolerances for wired glass

Thickness		Tolerance, thickness			
		Plus		Minus	
Inch	Mm.	Inch	Mm.	Inch	Mm.
7/32[a]	5.6	1/32	0.8	1/64	0.4
1/4	6.4	3/64	1.2	0	0
3/8	9.5	1/64	.4	3/64	1.2

a - Not approved by Underwriters' Laboratories for fire resistance.

3.5.1.1.2 Wire and mesh.

(a) Wire. Diameter of wire shall be not less than 0.020 inch (0.51mm.). Discoloration and slight distortion of wire is permissible.

(b) Mesh. Mesh m2, hexagonal shall be twisted. Meshes m1, m3, and m4 shall be welded. Spacing of parallel strands, mesh m5, shall be 1/2 ± 3/32 inch (12.7 ± 2.4mm.). Openings in meshes m1 to m4 inclusive shall not exceed 1-1/4 square inches (7.1 sq. cm.).

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3.5.1.1.3 Glass, polished or figured.

(a) Form 1, polished both sides. Glass may contain waviness which does not interfere with vision normal to the sufrace.
(b) Form 2, figured both sides. Glass shall not contain fire cracks, exposed wire or stones which can cause spontaneous breakage.

3.5.1.2 Form 3, figured glass.

3.5.1.2.1 Thickness and dimensional tolerances.

(a) Finishes f1 and f2, patterns p1 to p10. The allowable tolerances for cut rectangular shapes shall be plus or minus 1/16 inch (1.6mm.) for each 1/8 inch (3.2mm.) of thickness. Thickness and tolerances on thickness shall be in accordance with table VIII (see 6.2).

TABLE VIII. Thickness and tolerance for figured glass, patterns p1 to p10

Thickness		Tolerance, thickness			
		Plus		Minus	
Inch	Mm.	Inch	Mm.	Inch	Mm.
1/8	3.2	3/64	1.2	1/64	0.4
3/16	4.8	1/32	.8	1/64	.4
7/32	5.6	3/64	1.2	1/64	.4
1/4	6.4	3/64	1.2	1/64	.4
5/16	7.9	3/64	1.2	1/64	.8
3/8	9.5	3/64	1.2	1/32	.8

3.5.1.2.2 Glass. Figured glass shall be free from fire cracks and stones which can cause spontaneous breakage.

3.5.1.2.3 Surface pattern.

(a) Quality q10, decorative. Surface pattern shall be clear, sharp, and defined, and free from disfiguration which affects the appearance of the pattern. The smooth side shall be free from open gaseous inclusions greater than 3/32 inch (2.4mm.) or equivalent elliptical open inclusions up to 5/8 inch (15.9mm.) long with no more than 3 of the maximum size in each 50 square feet (4.64m.^{2}) of each cut sheet.
(b) Quality q11, glazing. Surface pattern shall be free from large areas of blemishes. Scattered areas of nonuniform surface and scattered surface blemishes are permissible. The smooth side shall be free from imbedded foreign materials, and shall be free from open gaseous inclusions greater than 1/8 inch (3.2mm.) or equivalent elliptical open inclusions up to 3/4 inch (19mm.) long with no more than 3 of the maximum size in each 50 square feet (4.64m.^{2}) of each cut sheet.

3.5.1.3 Form 4, rough surface finishes (rolled rough plate).

3.5.1.3.1 Thickness and dimensional tolerances. Allowable tolerance on all thicknesses of rough surfaced glass shall be plus 3/64 inch (1.2mm.) minus 1/32 inch (0.8mm.). Tolerances on lengths and widths shall be in accordance with table IX (see 6.2).

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TABLE IX. Thickness and tolerances for rough surfaced glass

Finish	Thickness		Tolerance, length and width, cut sizes plus or minus	
	Inch	Mm.	Inch	Mm.
f3, and f4 polished or smooth, one side, rough one side	7/32	5.6	1/16	1.6
	17/64	6.7	1/16	1.6
	21/64	8.3	1/16	1.6
	25/64	9.9	1/16	1.6
	33/64	13.1	3/32	2.4
f5, rough both sides	15/64	6.0	1/16	1.6
	9/32	7.1	1/16	1.6
	11/32	8.7	1/16	1.6
	13/32	10.3	1/16	1.6
	17/32	13.5	1/8	3.2

3.5.1.3.2 Glass, rough surfaced glass. Rough surfaces, finishes f3 and f5, shall not have open gaseous inclusions larger than 1/16 inch (1.6mm.) long heavy scratches, heavy rubs, smoke, or surface chips larger than 1/4 inch (6.4mm.) by 1/64 inch (0.4mm.) deep (see note(g), table III). The polished side of finish f3 shall meet the requirements for quality q3 plate glass in table III.

3.5.2 Kind B, classes 1, 2 and 3.

3.5.2.1 Dimensions and tolerances, forms 5 and 6. Dimensions and tolerances shall be as follows (see 6.2):

Length - 144 inches (3.66m.), maximum

Width

Form 5 - 25 and 47-1/2 or 50 inches.
(63.5 and 121 or 127cm.)

Form 6 - 27-1/2 inches (67.8cm.)

Thickness - 3/8 inch (9.5mm.)

Corrugations - Overall height 1 inch (2.5cm.), and pitch 2.50 inches (6.4cm.)

Tolerances:

	Plus or minus	
	Inch	Mm.
Length	3/16	4.7
Width	3/16	4.7
Thickness	3/64	1.2

Corrugations:

Height	1/8	3.2
Pitch	3/16	4.7

3.5.2.2 Plain and wired, forms 5 and 6. Glass shall be as specified in 3.5.1.1.3.

3.5.2.3 Wire and mesh, form 6. Mesh shall be hexagonal as specified in 3.5.1.1.2. Wire shall also be as specified in 3.5.1.1.2.

3.6 Class 2, types I, II, and III heat absorbing.

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3.6.1 Transmittance, styles A and B. The illuminant C (daylight), and solar transmittance shall be in accordance with table X (see 4.4.2, 4.4.3, and 6.2).

TABLE X. Transmittance for heat absorbing glass

Thickness, styles A and B	ICI illuminant C transmittance[a]		Solar transmittance[a] styles A and B
	Style A	Style B	
Inch	Percent, min.	Percent, max.	Total, percent, max.
1/8	80	80	65
1/4	70	70	50
3/8	60	60	40
1/2	50	50	35

(a) Straight line in interpolation may be used for intermediate thicknesses as an approximation.

3.6.2 Thickness and dimensional tolerances. Thickness and tolerances shall be according to the requirement already specified in the applicable glass.

3.6.3 Defects. The type and number of defects shall be not greater than that already specified for the applicable glass with the addition that light smoke is acceptable.

3.6.4 Tint. The tint of each type, style, kind, and form shall be as specified (see 6.2 and 6.5).

3.7 Class 3, types I, II and III, light reducing, tinted.

3.7.1 Thickness and dimensional tolerances. Thickness and tolerances shall be according to the requirements already specified for the applicable glass.

3.7.2 Defects. The type and number of defects shall be not greater than that already specified for the applicable glass with the addition that light smoke is acceptable.

3.7.3 Tint. The tint of each type, kind, and form shall be as specified (see 6.2 and 6.5).

3.7.4 Luminous transmittance. Light reducing glass is produced in a wide range of luminous transmittance values. The transmittance of each type and form of glass shall be as specified (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type, class, style, kind, form, and quality of glass offered for delivery at the same time.

4.2.2 Visual examination of end item. Sampling for visual examination shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Samples for tests. The number of samples selected for tests shall be in accordance with level S-2 in MIL-STD-105.

4.3 Examination.

4.3.1 End item. Glass selected in accordance with 4.2.2 shall be examined for defects listed in table XI. The acceptable quality levels (AQL's) in MIL-STD-105 shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE XI. Classification of defects, end item

Defects	Major	Minor
Type, class, style, and form not as specified	X	
Edges not as specified	X	
Dimensions not within tolerance	X	
Tint of heat absorbing or light reducing glass not as specified	X	
Defects greater or larger than those permitted for the specified glass	X	
Waviness excessive	X	
Labels on glass missing when required		X
Damage or defects affecting function or serviceability	X	
Damage or defects not affecting function or serviceability		X

4.3.2 Wave, ream, strings and lines, types I and II. Place sample in a vertical position at a distance of approximately 36 inches (91.5cm.) from a brick wall or similar background showing essentially straight lines. The viewer shall look through the sample at a distance of approximately 36 inches (91.5cm.) from the sample using daylight without direct sunlight or with background light suitable for observing each type of defect. View the sample at an angle to the surface of not less than the vision interference angle in tables II, III, V, and VI for the applicable glass. Line of vision shall be perpendicular to wall.

4.3.3 Scratches, rubs, stones, and gaseous inclusions, types I and II. Place samples in a vertical position approximately 36 inches (91.5cm.) from the viewers position. The viewer shall look through the sample using daylight without direct sunlight or with background light suitable for observing each type of defect.

4.3.4 Cut sizes. Measure length and width from edge to edge including flares.

4.3.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packing and marking comply with the requirements in section 5. Defects shall be scored in accordance with table XII. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6.5 defects per hundred units.

TABLE XII. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

4.4 Test methods.

4.4.1 Specimens. Select two specimens, one each from two samples selected in 4.2.3. For samples with figured surfaces, grind and polish surfaces to a smooth surface.

4.4.2 Luminous transmittance. Measure transmittance measure by illuminating each specimen with light having the spectral composition of ICI illuminant "C" (see 3.6.1 and 6.3). Measure the ratio of transmitted to incident luminous flux by one of the following methods:

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(a) Visual photometer.
(b) Corrected barrier-layer cell such as a Weston Photronic cell equipped with a Viscor filter.
(c) Calculation from the spectral distribution of illuminant "C" (see 6.3).

4.4.3 Total solar transmittance. Calculate transmittance for solar energy from the measured spectral transmittance of the glass, using the values for spectral distribution of solar energy received at sea level by a surface perpendicular to the sun's rays through an air mass of 2 (see 3.6.1 and 6.4).

4.5 Rejection.

4.5.1 Tests. A lot shall be rejected for failure to meet the test requirements of this specification when tested in accordance with 4.4.

4.5.2 Retests. Conduct retests using 2 specimens.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Class of like description in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style B, or PPP-B-621, class 2. Each pane of glass shall be separated with a sheet of neutral or slightly acid newsprint or kraft paper of equal dimensions to the glass, plus or minus 1/4 inch, or wood flour. The boxes shall be lined on all interior surfaces with not less than three layers of fiberboard conforming to PPP-F-320, type CF, class domestic, variety SW. The box shall be closed and strapped in accordance with the appendix to the applicable box specification, except that banding shall conform to type I, class B of QQ-S-781. When the gross weight of the box exceeds 200 pounds, skids shall be provided as specified in the applicable box specification.

5.1.1.1 Alternative bid. If bidders desire to use their standard method of export packing for glass and consider same of equal, or superior to the above requirements, they may submit alternative proposals based on such standard method of export packing. In case such alternative proposals are submitted, bidder should state in detail, the method of packing proposed, such statement to be written on a separate slip. The Government reserves the right to reject such alternative bids if it does not consider the method of packing satisfactory.

5.1.2 Level B. The glass shall be packed as specified in 5.1.1, except that containers shall be of a domestic type. Fiberboard containers conforming to PPP-B-636 class weather-resistant and PPP-B-640, class 2 may also be utilized.

5.1.3 Level C. The glass shall be packed to insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Plate and float.

(a) Silvering quality. Intended for silvering for mirror applications and is seldom used for glazing.
(b) Mirror glazing quality. Intended for mirror applications.
(c) Glazing quality. Intended for glazing windows and doors and other uses such as furniture, shelves and portlights.

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6.1.2 Sheet.

(a) AA quality. Intended for uses where superior quality is required.
(b) A quality. Intended for selected glazing.
(c) B quality. Intended for general glazing and other uses.
(d) A silvering quality. Intended for silvering for mirror applications and is seldom used for glazing.
(e) B silvering quality. Intended for mirror applications and is seldom used for glazing.
(f) Greenhouse quality. Intended for greenhouse glazing or similar applications where quality is unimportant.

6.1.3 Rolled.

(a) Decorative quality. Intended for use where design and esthetic characteristics are major considerations.
(b) Glazing quality. Intended for general glazing where functional or esthetic characteristics are a consideration.
(c) Wired glass. Intended for skylights and general glazing where fire retardation, security and safety are a consideration.
(d) Corrugated glass. Intended for structural applications including partitions, sidwwalks and window areas.

6.1.4 Heat absorbing. Intended for glazing where reduction of solarheat is required.

6.1.5 Light reducing. Intended for reducing the amount of visible light transmission for use in applications where glare is a problem.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, kind, form, and quality (see 1.2, 6.5 and 6.6).
(c) Edges (see 3.2.1).
(d) Marking (see 3.2.2).
(e) Plate or float glass, thickness (see 3.3.1).
(f) Sheet glass, designation or thickness (see 3.4.1).
(g) Wired glass, thickness (see 3.5.1.1 and 6.8).
(h) Corrugated glass, form 5, width (see 3.5.2.1).
(i) Figured, whether pattern should be on one side or both sides (see 1.2) and thickness (see 3.5.1.2.1).
(j) Rolled rough plate glass, thickness (see 3.5.1.3.1 and table IX).
(k) Heat absorbing glass, thickness (see 3.6.1, 3.6.2 and table X), and tint (see 3.6.4 and 6.5).
(l) Light reducing glass, thickness, form and transmittance (see 3.7.1 and 3.7.4), and tint (see 3.7.3 and 6.5).
(m) Size of lot, if different from 4.2.1.
(n) Selection of applicable level of packing required (see 5.1).
(o) Quantities of glass required in shipping container (see 5.1.1).

6.3 Luminous transmittance. Detailed information on the ICI illuminant "C" is contained in the following reference:

Journal of the Optical Society of America, Volume 34, page 635, 1944.

Spectral curve based on standard ICI luminoisity function is contained in each of the following references:

Proceedings, Sixth Session International Commission on Illumination, Geneva, page 67, 1924.

Journal of the Optical Society of America, Volume 34, page 641, 1944.

National Bureau of Standards, Journal of Research, Volume 6, RPZ89, page 465, 1931.

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6.4 Radiant transmittance. Detailed information on the measurement of radiant transmittance is given by Parry Moon in "Proposed Standard Solar - Radiation Curves for Engineering Use" of Journal Franklin Institute, Volume 230, page 604, table III, 1940.

6.5 Tint. Heat tints absorbing glasses and light reducing glasses are available in a variety of tints. These types of glass vary in tint between different manufacturers and from melt to melt so some variation in tint may occur.

6.6 Rolled pattern, wired, and shaped glass. When ordering special rolled glass consideration should be given to wire configuration, and to pattern, surface texture, shape,or design.

6.7 Scratches and surface defects. In normal handling of glass slight scratches and other surface defects may occur. These may ordinarily be removed by buffing to allow the glass to comply with the specification.

6.8 Samples. If for any particular purpose samples with bids are necessary, such as wired glass which may contain numerous gaseous inclusions along the wire, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated.

6.9 Glossary of terms.

6.9.1 Glass and types of glass.

(a) Glass. Glass is an inorganic product of fusion which has cooled to a rigid condition without crystallizing. Glass is typically hard and brittle, and has a conchoidal fracture. It may be colorless or tinted, and transparent to opaque. Masses or bodies of glass may be tinted, translucent, or opaque by the presence of dissolved, amorphous, or crystalline material. Glass that does not contain such added materials is designated as "clear" (transparent, class I) in this specification, even though the finished product may not be transparent in the ordinary sense as a result of the pattern (wired, corrugated, figured) furnished.
(b) Plate glass. Glass from which surface irregularities have been removed by grinding and polishing, so that the surfaces are plane (flat) and parallel.
(c) Float glass. Glass which has virtually plane and parallel surfaces formed by floating in a continuous ribbon of glass on the surface of a both of molten metal in a controlled atmosphere.
(d) Sheet glass. Transparent, flat glass having glossy, fire-finished, apparently plane and smooth surfaces, but having a characteristics waviness of surface.
(e) Wired glass. Rolled glass having a layer of meshed wire completely imbedded in the sheet.
(f) Corrugated glass. Glass rolled to produce a corrugated contour.
(g) Figured glass. Flat glass having a pattern on one or both surfaces.
(h) Heat-absorbing glass. Glass for intercepting appreciable portions of radiant energy-especially solar energy.
(i) Light-reducing glass. Glass which is formulated to reduce the transmission of light.

6.9.2 Defects in glass.

(a) Crush. A lightly pitted area resulting in a dull gray appearance over the region.
(b) Digs. Deep, short scratches.
(c) Dirt. A small particle of foreign matter imbedded in the glass surface.
(d) Gaseous inclusions. Round or elongated bubbles in the glass.
(e) Knot. A transparent area of incompletely assimilated glass having an irregular knotty or tangled appearance.
(f) Lines. Fine cords or strings, usually on the surface of sheet glass.
(g) Open gaseous inclusions. Bubbles at the surface of glass which are open leaving a cavity in the finished surface.
(h) Process surface defects. The surfaces of plate glass having very fine surface defects remaining from the grinding and polishing process consisting of fine pits and cracks which are denoted as "finish". When this condition is visible it is called "short finish". Float glass can also have some slight surface defects which originate in the process. These can be small particles of foreign materials on either surface or slight defects in the bottom (float)surface.

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(i) Ream. Inclusions within the glass or layers or strings of glass which are not homogeneous with the main body of the glass.
(j) Rubs. Abrasion of the glass surfaces producing a frosted appearance. A rub differs from a scratch in having appreciable width.
(k) Scratches. Any marking or tearing of the surface produced in manufacturing or handling, appearing as though it were done by a sharp or rough instrument.
(l) Smoke. Streaked areas appearing as slight discoloration.
(m) Stones. Any crystalline inclusion imbedded in the glass.
(n) Strings. Transparent lines appearing as though a thread of glass had been incorporated into the sheets.
(o) Wave. Defects resulting from irregularities of the surfaces of glass making objects viewed at varying angles appear wavy or bent.

6.10 The metric equivalents to dimensional requirements are for information only, and are not intended to be used for determining acceptance or rejection of items otherwise conforming to the requirements of the specification.

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review:

Army - ME, CE
Navy - YD
Air Force - 84

User:

Navy - MC

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